

AI-Enabled Layout Automation for Analog and RF IC: Current Status and Future Directions

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Abstract—This paper provides an overview of AI-enabled layout automation for analog and RF integrated circuits (ICs). We examine recent advancements in machine learning algorithms, optimization techniques, and AI-driven design methodologies tailored to the unique challenges of analog and RF IC layout. The paper highlights the benefits, limitations, and key achievements of these AI approaches and identifies existing gaps. Finally, we discuss future directions and research opportunities, emphasizing AI's potential to enhance the efficiency and accuracy of IC layout design.

I. CURRENT STATUS

Analog and RF integrated circuits (ICs) are crucial components in modern electronic systems, requiring meticulous design and layout to ensure optimal performance. Traditional manual design processes for these circuits are often labor-intensive and time-consuming, limiting the ability to explore a wide design space and leading to suboptimal solutions. As the demand for high-performance analog and RF ICs increases, there is a growing need for efficient, automated layout tools that can enhance design productivity and allow for comprehensive design space exploration.

A. Analog IC Layout Automation

There have been significant developments in automatic analog layout generation using AI techniques. By leveraging end-to-end analog layout generation tools, such as MAGICAL [1], [2] and ALIGN [3], researchers have access to real-world and verifiable layouts.

Particularly, AI techniques are used in four subproblems of analog IC generation. 1) **Geometric Constraint Generation**: Geometric constraints are crucial for ensuring high performance and yield in analog designs, particularly under process-voltage-temperature (PVT) variations. Symmetry constraints, often essential in differential topologies, help reduce mismatch and improve performance [4]–[12]. 2) **Placement**: The placement stage determines the locations of modules and devices, considering constraints to mitigate layout effects on electrical performance. ML techniques have been used in modeling wells [13], [14] and performance [15]–[17] in the analog placement stage. 3) **Routing**: Routing in analog layout design involves connecting components with metal wires while meeting performance, electrical, and geometric constraints to ensure optimal functionality and manufacturability of the

circuit. Geniusroute [18] uses a variational autoencoder to learn manual routing strategies and applies them in automatic detailed routing. Chen et al. [19] propose a reinforcement learning (RL)-based routing approach that aids in detailed routing by optimizing routing patterns through trial-and-error learning, allowing the algorithm to explore and exploit various routing strategies to meet complex constraints and performance criteria. Xu et al. [20] propose to directly model analog performance using a 3DGNN model. The model generates guidance for an automatic detailed router, thus enabling a performance-driven analog routing paradigm. A detailed review on AI for analog IC layout automation can be found in [21]. A recent trend is the use of performance-driven analog IC layout optimization leveraging machine learning models. A related survey can be found in [22].

B. RF IC Layout Automation

Compared to analog IC layouts, automatic RF layout generation is relatively less studied. RF ICs consist of some components whose layouts are similar to those of analog ICs, while others introduce new challenges.

Some components are similar to those in analog ICs, therefore it is possible to utilize an analog IC layout generator to synthesize the layout. Poojaray et al. [23] propose using ALIGN [3] to generate MIMO receiver designs. With automatically generated MIMO receivers, users can easily explore the design space, thus improving efficiency. Wei et al. [24] develop a PCELL generator for RF devices. The device layouts are further integrated with commercial EDA tools.

The layout synthesis of passive devices in RF has a high degree of freedom, resulting in a more complicated optimization problem. Usually, in RF IC layout design, evaluating a passive device layout requires electromagnetic (EM) simulation to determine the EM behavior. As the geometry can be arbitrary except for manufacturing constraints, the geometry of passive devices is much more flexible than that of typical analog PCELLs. Lyu et al. [25] incorporate passive device layout generation in a parameter sizing framework. By representing inductor and transformer devices with several parameters, a Bayesian optimization framework optimizes the layout geometries for higher performance. Similar approaches are also adopted in [26], [27].

On the other hand, some devices are more sensitive to layout geometries, resulting in challenging EM simulation

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and geometric optimization. Karahan et al. [28] treat layout generation as an inverse problem. Their work uses a deep convolutional neural network surrogate model to simulate the EM process. It takes the arbitrary planar EM structure geometry as input and predicts its scattering parameters. Then, the geometry can be further optimized by solving the inverse problem of this model.

II. PERSPECTIVES AND FUTURE DIRECTIONS

The landscape of AI-enabled layout automation for analog and RF ICs is rapidly evolving, driven by advancements in machine learning (ML) and artificial intelligence (AI) technologies. Several future directions can significantly enhance the efficiency, accuracy, and capabilities of automated layout tools.

Foundational Circuit Representation Model: A critical future direction is the development of foundational models for circuit representation [29]. These models, based on advanced ML techniques such as graph neural networks (GNNs), can accurately capture the hierarchical and topological features of circuit designs. By understanding the detailed structure and constraints of analog and RF circuits, these models can significantly enhance the recognition and optimization of circuit blocks. GNNs, for example, can perform subcircuit annotation and symmetry constraint extraction by processing the graph representations of circuits. Additionally, these models can integrate reinforcement learning to dynamically optimize placement and routing decisions and generative adversarial networks (GANs) to explore diverse layout configurations, thereby broadening the design space.

Generative AI and Large Language Models (LLMs): Generative AI and LLMs present novel opportunities for enhancing layout automation. LLMs, such as GPT-4, can assist in generating design documentation, providing detailed explanations of design choices, and even suggesting optimizations based on historical data and design patterns. Generative AI models can create new design layouts by learning from vast amounts of existing design data, thus providing innovative solutions that might not be immediately apparent to human designers. These models can also help in predicting the performance of new layouts, reducing the need for extensive simulations. Generative model has shown that a machine learning model can inspect human designs and transfer the relevant knowledge into automatic designs [13], [14], [18]. A recent work further shows a LLM model can assist to find the layout constraints for analog layouts [30].

Human-in-the-Loop Systems: While the ultimate goal is fully automated layout generation, incorporating human expertise can significantly enhance the quality and reliability of automated tools. Future systems can be designed to allow for human intervention at critical stages, where designers can provide insights or override automated decisions. This approach combines the efficiency of automation with the expertise of human designers, leading to superior design outcomes. Techniques such as active learning, where the

AI system queries human experts for guidance on uncertain decisions, can be particularly effective in this context.

Real-Time Feedback and Iterative Improvement: Incorporating real-time feedback mechanisms can significantly enhance the iterative design process. Future tools can provide designers with instant feedback on the impact of their design choices, allowing for quick adjustments and optimizations. Additionally, iterative improvement algorithms can continuously refine the layout based on performance simulations and measurements, ensuring that the final design meets all specifications. Techniques such as Bayesian optimization can be used to systematically explore the design space and identify optimal configurations efficiently.

III. CONCLUSION

AI-enabled layout automation for analog and RF ICs is rapidly transforming the landscape of circuit design, addressing the complexities and inefficiencies of traditional manual methods. By integrating advanced ML techniques, such as GNNs, GANs, and reinforcement learning, these tools can now automate placement and routing with a level of precision and speed previously unattainable. The incorporation of generative AI and large language models further enhances the design process, offering innovative solutions and detailed documentation support. As these technologies continue to evolve, they promise to significantly improve design productivity, optimize performance, and broaden the exploration of the design space, paving the way for more sophisticated and high-performing analog and RF ICs in the future.

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